



Let X be a $K3$ surface such that the lattice S_X is of rank 11 with the Gram matrix

$$\text{diag}(2, -2, -2, \dots, -2),$$

and the period $\omega_X \in T_X \otimes \mathbb{C}$ is general so that

$$\text{O}(T_X, \omega_X) = \{\pm \text{id}\}.$$

The lattice S_X is isomorphic to $I_{1,10}(2)$. Let $e_0, e_1, \dots, e_{10}$ be the vectors of $I_{1,10}$ such that

$$\langle e_0, e_0 \rangle = 1, \quad \langle e_i, e_i \rangle = -1 \quad (i = 1, \dots, 10), \quad \langle e_i, e_j \rangle = 0 \quad (i \neq j).$$

By Vinberg algorithm, we see that $I_{1,10}$ is generated by the vectors

$$\begin{aligned} \alpha_0 &= e_0 - e_1 - e_2 - e_3, \\ \alpha_1 &= e_1 - e_2, \\ &\dots \\ \alpha_9 &= e_9 - e_{10}, \\ \alpha_{10} &= e_{10}, \\ \alpha_{11} &= 3e_0 - e_1 - \dots - e_{10}, \end{aligned}$$

of norms $-2, -2, \dots, -2, -1, -1$, and $\text{O}(I_{1,10}, \mathcal{P})$ is generated by the reflections with respect to these vectors. Hence $\text{O}(S_X, \mathcal{P})$ is also generated by these 12 reflections. The order of $\text{O}(q_S)$ is 46998591897600. The subgroup of $\text{O}(q_S)$ generated by the image of these 12 reflections by the natural homomorphism $\eta_S: \text{O}(S_X, \mathcal{P}) \rightarrow \text{O}(q_S)$ is also equal to 46998591897600. Hence η_S is surjective, and the kernel of η_S is of index 46998591897600 in $\text{O}(S_X, \mathcal{P})$. Note that the discriminant group of S_X is 2-elementary, and hence the period condition for $g \in \text{O}(S_X, \mathcal{P})$ is $\eta_S(g) = \text{id}$. Therefore we see that the image of

$$\text{Aut}(X) \rightarrow \text{O}(S_X, \mathcal{P})$$

is equal to the subgroup of the kernel of η_S consisting of elements that preserve the nef-and-big cone N_X .